

Precalculus Review (Part II) Homework

Name
Date
Course

Simplify.

① $\frac{\frac{3}{7}}{\frac{5}{6}}$

② $\frac{\frac{4}{5}}{7}$

③ $\frac{1}{\frac{8}{9}}$

Determine if the following values can be discovered without a calculator. Then find the values. If a calculator is necessary, round your answer to the nearest thousandth.

④ $\log_3 9$

⑤ $\log_6 8$

⑥ $\log_7 1$

⑦ $\ln 4$

⑧ $\log 2$

⑨ $\ln e^2$

⑩ $\log 100$

Use the graphs below to find the following values.

⑪ $f(1)$

⑫ $f(-2)$

⑮ $f(-5)$

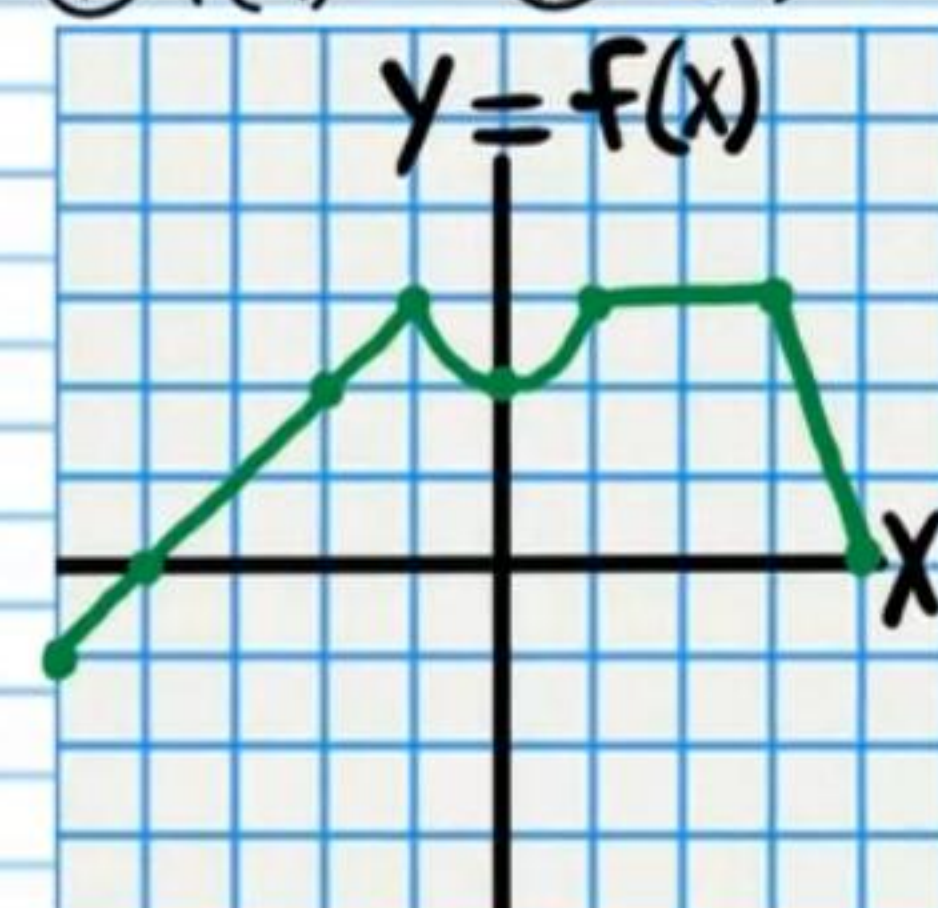
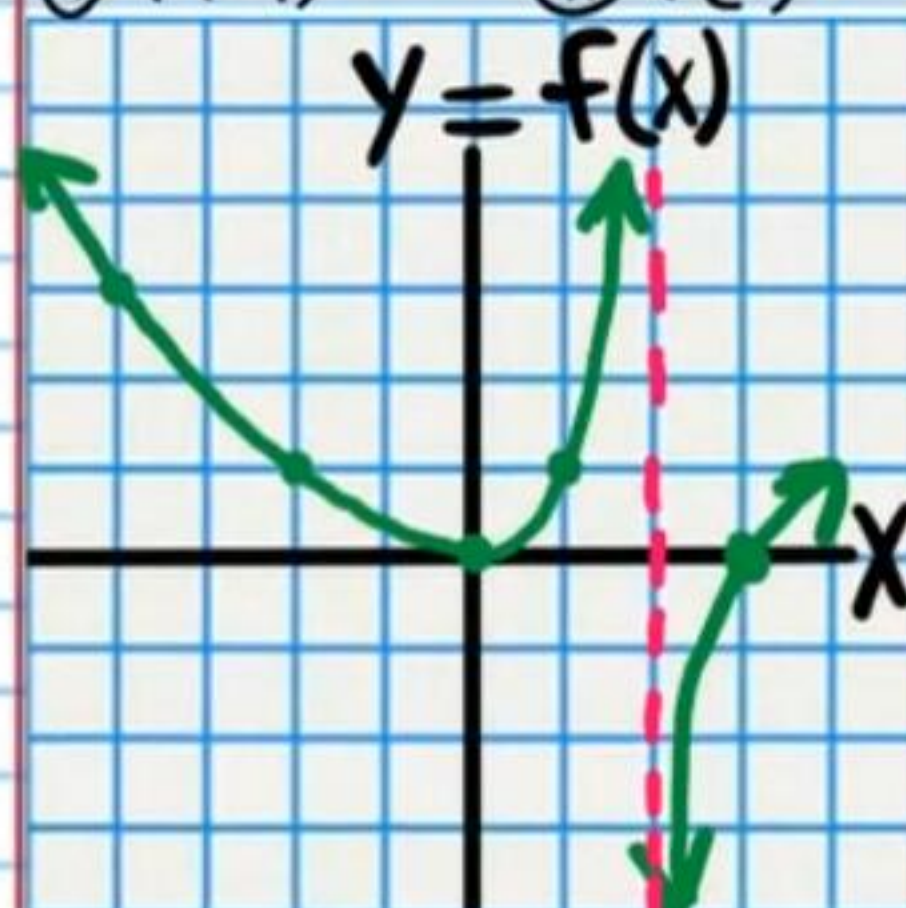
⑯ $f(4)$

⑬ $f(-4)$

⑭ $f(0)$

⑰ $f(-1)$

⑱ $f(-3)$



Use the piecewise functions to find the values of the expressions.

$$(19) f(x) = \begin{cases} 4^x, & -1 \leq x \leq 1 \\ \ln x, & 2 < x < 4 \end{cases}$$

a) $f(0)$

b) $f(3)$

c) $f(-1)$

d) $f(e)$

$$(20) f(x) = \begin{cases} x^2 + 3x + 2, & x < 5 \\ \frac{2}{x}, & x \geq 5 \end{cases}$$

a) $f(1)$

b) $f(-2)$

c) $f(5)$

d) $f(6)$

$$(21) f(x) = \begin{cases} 10x, & -1 > x \\ -\pi, & -1 \leq x \end{cases}$$

a) $f(0)$

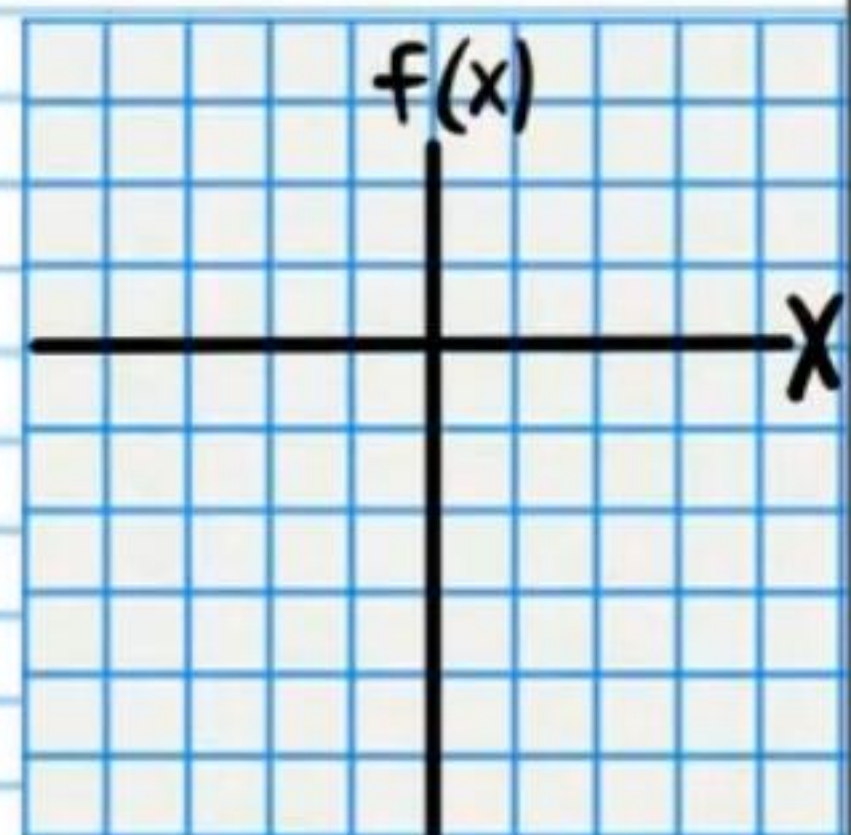
b) $f(-1)$

c) $f(-6)$

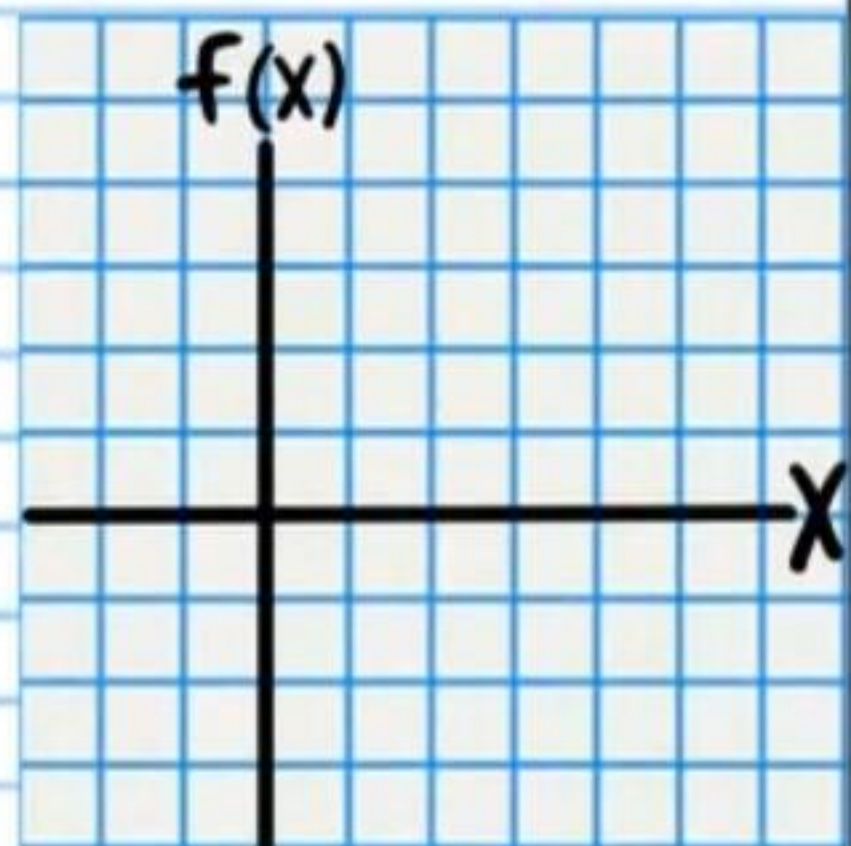
d) $f(-1.5)$

Graph the following functions.

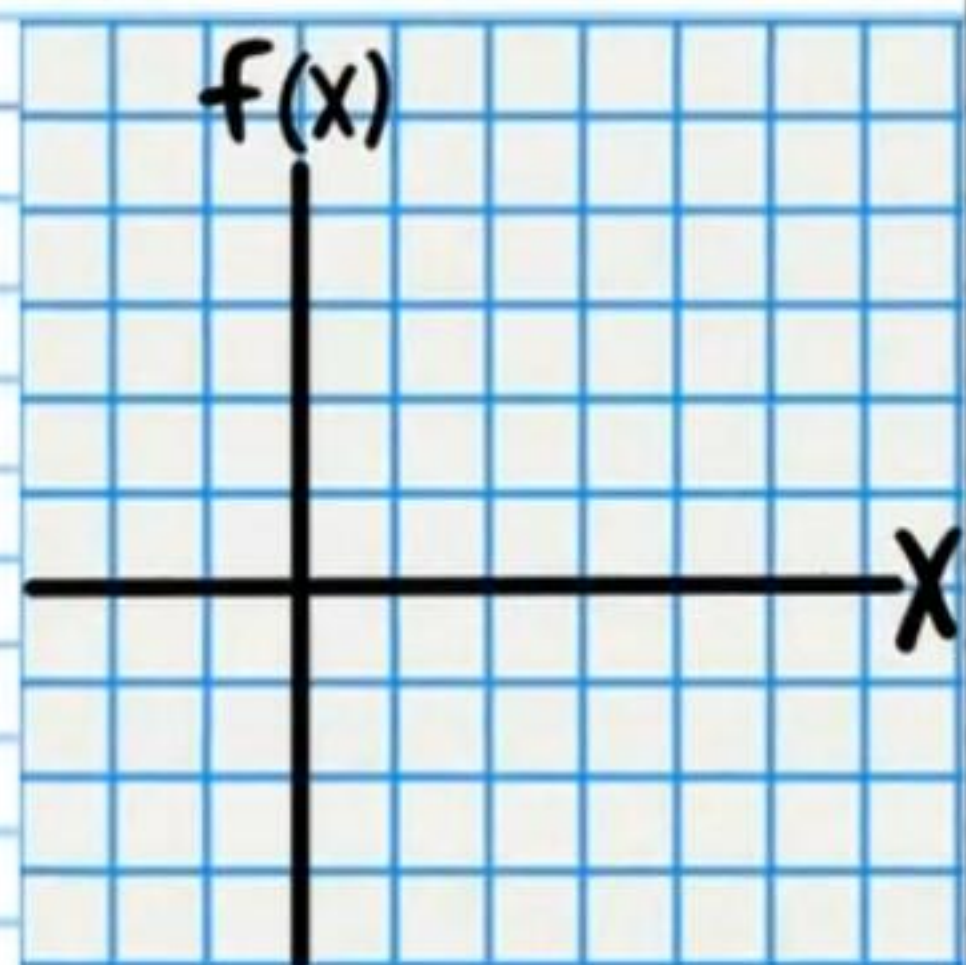
$$(22) f(x) = -x^2$$



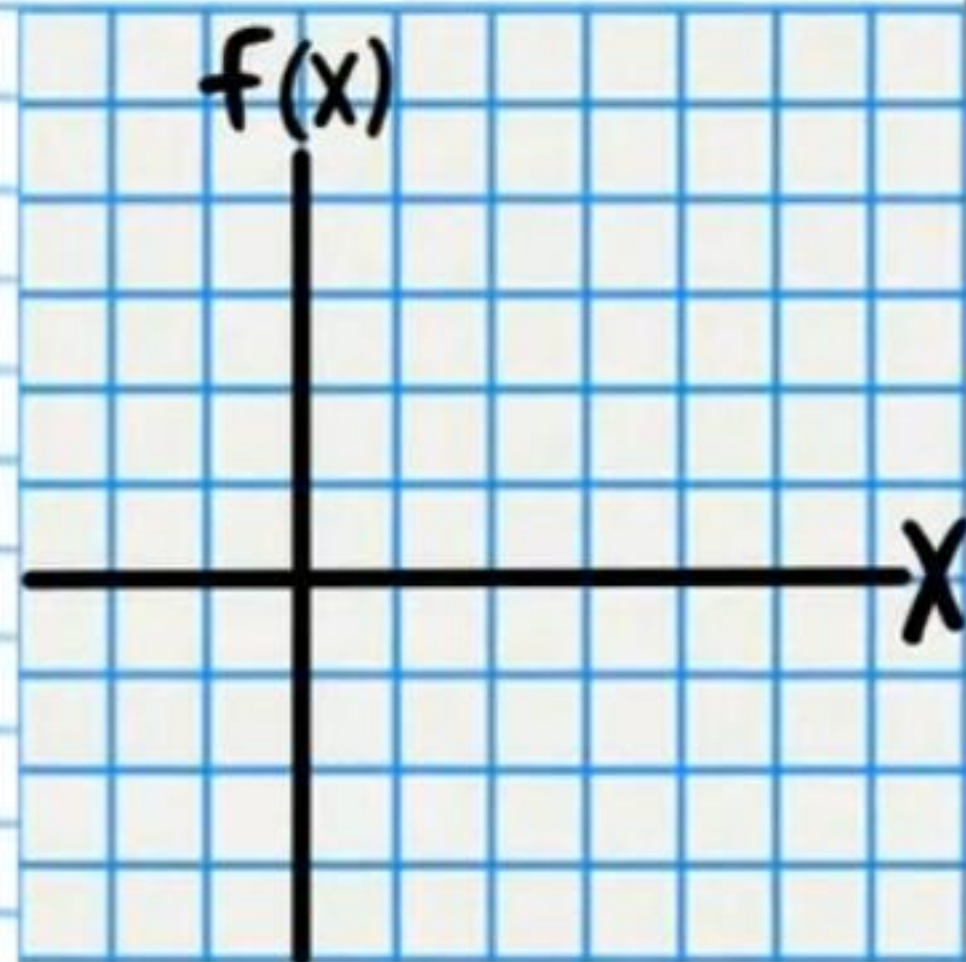
$$(23) f(x) = \sqrt{\frac{x}{3}} - 1$$



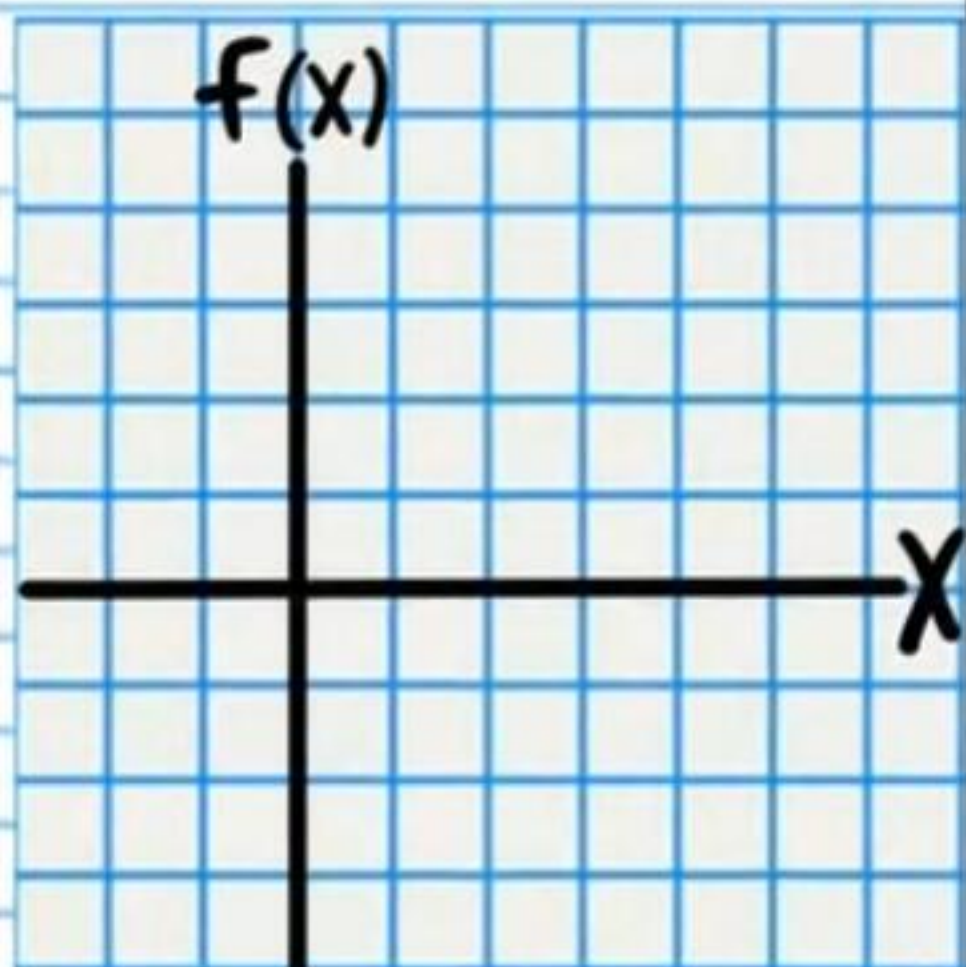
②4 $f(x) = -(2x-4)^3$



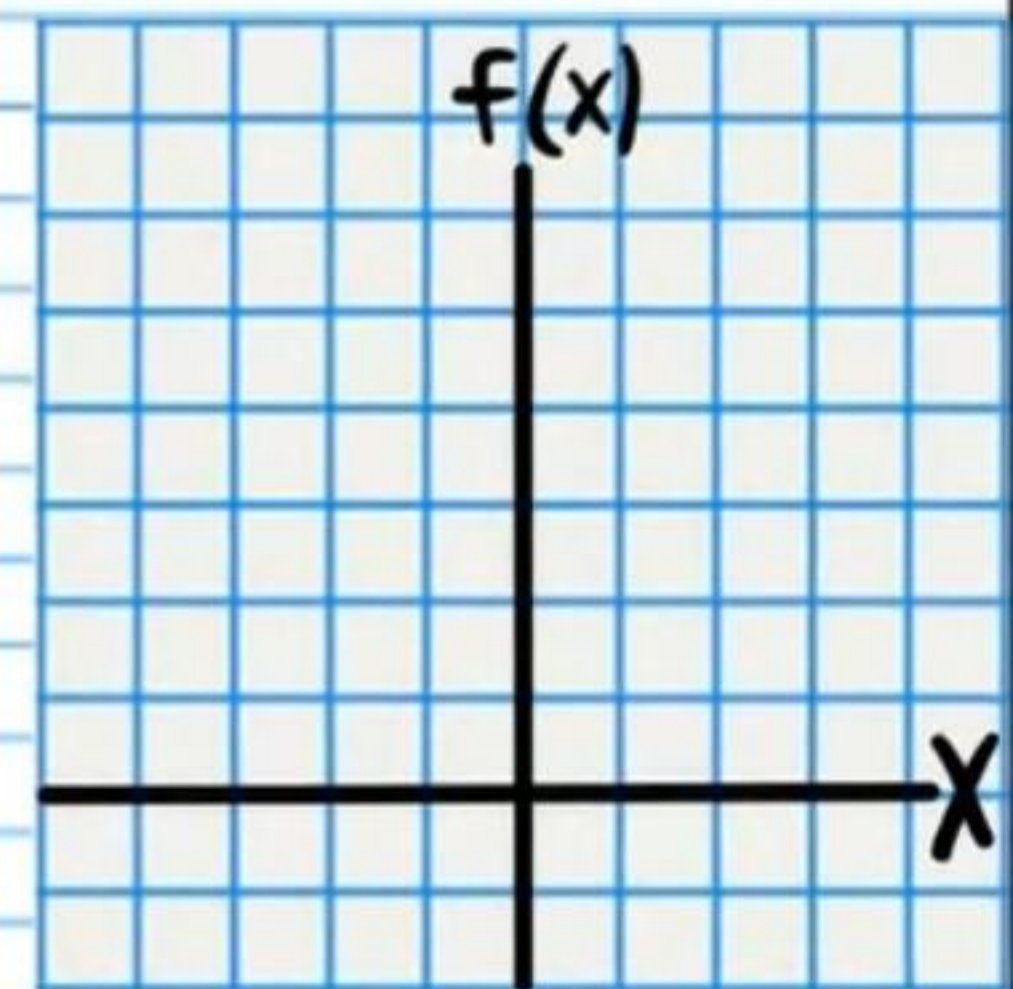
②5 $f(x) = \log_3(x-2)$



②6 $f(x) = 2(3^x) - 1$



②7 $f(x) = |-3(x+2)| + 2$



Answers

① $\frac{18}{35}$

② $\frac{4}{35}$

③ $\frac{9}{8}$

④ No Calculator
2

⑤ Calculator
1.161

⑥ No Calculator
0

⑦ Calculator
1.386

⑧ Calculator
0.301

⑨ No Calculator
2

⑩ No Calculator
2

⑪ 1

⑫ 1

⑬ 3

⑭ 0

⑮ -1

⑯ 0

⑰ 3

⑱ 1

⑲ a) 1 b) $\ln 3$ c) $\frac{1}{4}$ d) 1

⑳ a) 6 b) 0 c) $\frac{2}{5}$ d) $\frac{1}{3}$

㉑ a) $-\pi$ b) $-\pi$ c) -60 d) -15

